PAGE NO :
DATE :

26/3/2026.

Assignment :

I] K-Means Clustering:

K-Means is an superised learning algorithm used to group data into K clusters based on similarity

* When to use :
  - → when no. of clusters is known
  - → Large dataset.
  - → Customer segmentation

* Why to use
  - → Simple & fast.
  - → Works well for spherical clusters.

* Working algorithm :
  - → Choose no. of clusters K.
  - → Initialize K centroids randomly.
  - → Assign each point to nearest centroid
  - → Update centroids.
  - → Repeat until centroids don't change.

* Mathematical Formula :

→ $J = \sum i = 1K \sum x \in C_i \|x - \mu_i\|2.$

where, C_i = cluster

μ_i = centroid

$\|x - \mu\|$ = distance

Scanned with OKEN Scanner

PAGE NO :
DATE :

Example:

Points : (1,2), (2,3), (8,9), (9,10), $k = 2$.

Step 1 : Initial centroids → (1,2), (8,9).

Step 2 : Assign points

Cluster 1 → (1,2), (2,3)

Cluster 2 → (8,9), (9,10)

Step 3 : Update centroid

C1 → (1.5, 2.5)

C2 → (8.5, 9.5)

Repeat → clusters stabilize.

2] Hierarchical Clustering :

It builds a tree-like structure of clusters.

* Types:

→ Agglomerative (bottom-up)

→ Divisive (top-bottom)

* Working (Agglomerative) :

→ Start with each point as its own cluster

→ Compute distance b/w clusters

→ Merge closest clusters

→ Repeat until one cluster remains

* Distance Measures :

→ Single linkage ⇒ min distance

→ Compute linkage → max distance

→ Average linkage

Scanned with OKEN Scanner

PAGE NO :
DATE :

Example:

Points : A, B, C

Step 1: A, B, C separate

Step 2: Merge closest (A & B)

Step 3: Merge (AB) with C

Final → one cluster

3] DBSCAN (Density-Based Spatial Clustering of Applications with Noise):

DBSCAN groups points based on density.

* When to use:
  - → When clusters are irregular shape
  - → When noise / outliners present.

* Why to use:
  - → Detects outliers automatically
  - → No need to specify K.

* Key terms:
  - → Epsilon (ε): radius
  - → MinPts : minimum points
  - → Core point : dense region
  - → Noise : outliner

* Working Algorithm:
  - → Pick a point
  - → Find neighbors within ε
  - → If neighbors ≥ MinPts → from cluster
  - → Expand cluster
  - → Repeat for all points.

Scanned with OKEN Scanner

PAGE NO.
DATE :

* Importance
  - → works for non-linear shapes.
  - → Robust to noise
  - → Used ~~D~~ in anomaly detection.

4] GMM (Gaussian Mixture Model):

GMM is a probabilistic model that assumes data is generated from multiple Gaussian distribution.

* When to use:
  - → overlapping clusters
  - → Soft clustring needed.

* Why to use:
  - → Gives probability of belonging
  - → More flexible than K-Means.

* Working Algorithm:
  - → Assume K Gaussian distributions
  - → Calculate probability each point belongs to each cluster.
  - → Update mean, variance & weights
  - → Until convergence.

* Mathematical Formula:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

where, π_k = weight
μ_k = mean
Σ_k = variance

Scanned with OKEN Scanner

PAGE NO :
DATE :

Example:

Points may belong:

C1 → 70%

C2 → 30%

5] Evaluation Metrics for Clustering:

i] Silhouette Score:

Measures how similar a point is to its own ~~and~~ cluster vs others.

$$S = b - a / \max(a, b).$$

where, a = intra-cluster distance

b = nearest cluster distance

Range: -1 to 1.

Higher = better.

ii] Elbow Method:

Used to find optimal K

→ Plot: K vs error.

→ Choose point where curve bends.

iii] Davies-Bouldin ~~Index~~ Index:

→ Measures cluster similarity

→ Lower value = better.

iv] Calinski-Harabasz Index:

→ Ratio of b/w-cluster & within-cluster variance.

→ Higher = better.

Scanned with OKEN Scanner